5 Statistics

What students need to learn:

Ref	Content descriptor	Concepts and skills
SP a	Understand and use statistical problem solving process/handling data cycle	• Specify the problem and plan • Decide what data to and what statistical analysis is needed • Collect data from a variety of suitable primary and secondary sources • Use suitable data collection techniques • Process and represent the data • Interpret and discuss the data
SP b	Identify possible sources of bias	• **Discuss how data relates to a problem, identify possible sources of bias and plan to minimise it** • **Understand how different sample sizes may affect the reliability of conclusions drawn**
SP c	Design an experiment or survey	• Identify which primary data they need to collect and in what format, including grouped data • Consider fairness • Understand sample and population • Design a question for a questionnaire • Criticise questions for a questionnaire • **Design an experiment or survey** • **Select and justify a sampling scheme and a method to investigate a population, including random and stratified sampling** • **Use stratified sampling**
SP d	Design data-collection sheets distinguishing between different types of data	• Design and use data-collection sheets for grouped, discrete and continuous data • Collect data using various methods • Sort, classify and tabulate data and discrete or continuous quantitative data • Group discrete and continuous data into class intervals of equal width

Higher

Ref	Content descriptor	Concepts and skills
SP e	Extract data from printed tables and lists	• Extract data from lists and tables
SP f	Design and use two-way tables for discrete and grouped data	• Design and use two-way tables for discrete and grouped data • Use information provided to complete a two-way table
SP g	Produce charts and diagrams for various data types	• Produce: – Composite bar charts – Comparative and dual bar charts – Pie charts – Histograms with equal class intervals – Frequency diagrams for grouped discrete data – Scatter graphs – Line graphs – Frequency polygons for grouped data – **Grouped frequency tables for continuous data** – Ordered stem and leaf diagrams – **Cumulative frequency tables** – **Cumulative frequency graphs** – **Box plots from raw data and when given quartiles, median** – **Histograms from class intervals with unequal width** • **Use and understand frequency density**

Ref	Content descriptor	Concepts and skills
SP h	Calculate median, mean, range, **quartiles and interquartile range,** mode and modal class	• Calculate: – mean, – mode, – median, – range, – modal class, – the interval which contains the median • Estimate the mean of grouped data using the mid-interval value • Find the median, **quartiles and interquartile range** for large data sets with grouped data • Estimate the mean for large data sets with grouped data • Understand that the expression 'estimate' will be used where appropriate, when finding the mean of grouped data using mid-interval values • **Use cumulative frequency graphs to find median, quartiles and interquartile range** • **Interpret box plots to find median, quartiles, range and interquartile range**

Higher

Ref	Content descriptor	Concepts and skills
SP i	Interpret a wide range of graphs and diagrams and draw conclusions	• Interpret: – composite bar charts – comparative and dual bar charts – pie charts – stem and leaf diagrams – scatter graphs – frequency polygons – **box plots** – **cumulative frequency diagrams** – **histograms** • Recognise simple patterns, characteristics and relationships in line graphs and frequency polygons • **Find the median from a histogram or any other information from a histogram, such as the number of people in a given interval** • From line graphs, frequency polygons and frequency diagrams – read off frequency values – calculate total population – find greatest and least values • From pie charts: – find the total frequency – find the size of each category • Find the mode, median, range and **interquartile range**, **as well as the greatest and least values from stem and leaf diagrams** • **From cumulative frequency graphs:** – **estimate frequency greater/less than a given value** – **find the median and quartile values and interquartile range** • **From histograms:** – **complete a grouped frequency table** – **understand and define frequency density** (NB: No pictograms or bar charts at higher)

Ref	Content descriptor	Concepts and skills
SP j	Look at data to find patterns and exceptions	• Present findings from databases, tables and charts • Look at data to find patterns and exceptions • **Explain an isolated point on a scatter graph**
SP k	Recognise correlation and draw and/or use lines of best fit by eye, understanding what these represent	• Draw lines of best fit by eye, understanding what these represent • Distinguish between positive, negative and zero correlation using lines of best fit • Understand that correlation does not imply causality • Use a line of best fit, **or otherwise,** to predict values of one variable given values of the other variable • **Appreciate that correlation is a measure of the strength of the association between two variables and that zero correlation does not necessarily imply 'no relationship' but merely 'no linear relationship'**
SP l	Compare distributions and make inferences	• **Compare distributions and make inferences, using the shapes of distributions and measures of average and spread, including median and quartiles** • Compare the mean and range of two distributions, **or median and interquartile range**, as appropriate • Understand that the frequency represented by corresponding sectors in two pie charts is dependent upon the total populations represented by each of the pie charts • Use dual or comparative bar charts to compare distributions • Recognise the advantages and disadvantages between measures of average • **Compare the measures of spread between a pair of box plots/cumulative frequency graphs**

Higher

Ref	Content descriptor	Concepts and skills
SP u	Use calculators efficiently and effectively, including statistical functions	• Calculate the mean of a small data set, using the appropriate key or a scientific calculator • **Use $\sum x$ and $\sum fx$ or the calculation of the line of best fit**